Nicholas Verzic

NicholasVerzic@gmail.com
linkedin.com/in/nicholasverzic

EDUCATION

The University of Texas at Austin, Austin, TX
Master of Science: Artificial Intelligence — Expected Graduation: 08/27
Bachelor of Science: Mathematics; Minor: Economics — Completed: 05/25
Bachelor of Science: Mechanical Engineering — Completed: 05/25

SKILLS

Programming Languages/Software:
Advanced: Python, C++, JavaScript, Java, R, MATLAB, GNU Octave, SolidWorks, Unity, Office Suite, Adobe Suite
Intermediate: Hack, C#, Bash Scripting, C, Swift
Packages/Libraries: Robot Operating System, PyTorch, TensorFlow, PySpark, Numpy, Scikit-Learn, Keras, Pandas, Prometheus
Hardware and Manufacturing: Mechatronics, Raspberry Pi, Arduino, 3D Printing, Laser Cutting, Injection Molding, Soldering

INTERNSHIP/WORK EXPERIENCE

Meta (Menlo Park, CA)
Machine Learning Research Scientist — 10/25-Present
- Designing foundation ML model architectures for audio and video content understanding across Meta's Family of Apps
- Deploying content understanding ML models for copyright holder protection as part of Similarity Understanding team
- Architecting ML model architecture and feature optimization experiments, driving regular performance improvements on both fundamental research (TPR, FPR, AUC) and problem-specific metrics

Machine Learning Intern — 05/24-08/24
- Developed audio recognition ML model for determining types of audio transformations present in files, adapting vision transformer for use as audio classifier

Human-Enabled Robotic Technology Lab, The University of Texas at Austin (Austin, TX)
Undergraduate Researcher — 06/25-03/26
- Led research project constructing 6-degree-of-freedom wearable haptic sleeve for immersive surgical training
- Integrated custom VR environment with hand-tracked surgical tools, driving dynamic wearable forces simulating physical feedback experienced during chest tube insertion procedure
- Invited/Delivered Oral Presentation at 2026 IEEE Haptics Symposium (Reno, NV)

Hewlett Packard Enterprise (San Jose, CA)
Software Engineer Intern — 05/23-08/23
- Scripted custom machine and database monitoring tools to enable comparison of Aruba ClearPass Policy Manager versions
- Performed code repairs of different CPPM versions' daemon processes

Learning Agents Research Group, The University of Texas at Austin (Austin, TX)
Undergraduate Researcher — 06/22-08/25
- Led research project on novel technique for computer-vision detection and identification of elevator buttons/labels
- Invited/Delivered Oral Presentation at 2024 IEEE International Conference on Intelligent Robots and Systems (IROS) (Abu Dhabi, UAE)
- Developed spatial- and image-data processing scripts for live, accurate gaze detection pipeline given low-resolution video feed
- Assembled structure-from-motion pipeline for future environmental understanding projects

Applied Research Laboratories, The University of Texas at Austin (Austin, TX)
Student Technician — 06/22-08/22
- Studied propagation of nonlinear acoustic waveforms through special materials
- Wrote MATLAB scripts to calculate waveform data propagation over time via differential equations, applied to real experimental data

Stanford Institute for Human-Centered Artificial Intelligence's Digital Economy Lab, Stanford University (Stanford, CA)
Research Intern — 02/21-05/23
- Researched personal productivity trends, contributing to op-ed published to the MIT Technology Review titled "The coming productivity boom," under S-DEL Director, Professor Erik Brynjolfsson
- Programmed R scripts analyzing Californian demographic data for Stanford California 100 project

NASA Ames Research Center (Mountain View, CA)
NASA Intelligent Robotics Group Yearlong University Intern — 12/18-12/20
- Researched and developed custom multilateration algorithms for distributed sensor network using RF ranging in MATLAB
- Conducted on-site experiments in simulated lunar surface environment

INDEPENDENT PROJECTS/ACTIVITIES

IDBizz iOS Application 06/21-07/21

- Developed free iOS application for consumers to find and support business owners' identities and causes

ChildrenGiving.org Co-Founder, Lead Curriculum Designer 07/09-07/23

- Delivered free programming and CS workshops to 1,000+ students through Microsoft (Stanford) and Tracy School District

PUBLICATIONS

- IEEE Haptics Symposium, 2026 publication, "Improving Chest Tube Insertion Depth and Timing Error in an Immersive VR Training Task with a Wearable Haptic Sleeve"
- IEEE International Conference on Intelligent Robots and Systems (IROS), 2024 publication, "Recovering Missed Detections in an Elevator Button Segmentation Task"
- Acknowledgment in thesis, "Cross-Domain Adaptation and Geometric Data Synthesis for Near-Eye to Remote Gaze Tracking"
- IEEE Aerospace Conference, 2020 publication, "PHALANX: Expendable Projectile Sensor Networks for Planetary Exploration"

AWARDS & HONORS

- UT Austin University Honors recipient (all semesters)

UNIVERSITY EXTRACURRICULARS

- Autonomous Robotics Research Project Peer Mentor, Research Program Peer Mentor
- *University Clubs:* Design/Manufacturing Team Leader -- Longhorn Toy Factory, Active Guidance Team -- Texas Guadaloop